SAFETY PRECAUTIONS

WARNING - SHOCK AND BURN HAZARD:
Always disconnect power to the drive before removing or inserting a printed circuit card. Failure to do so may cause serious injury to personnel and damage to the drive or the driven machinery.

CAUTION: Treat all cards with static sensitive handling techniques. Use "grounding" strap when changing cards and store cards in the anti-static bags they are shipped in.

NOTE: *Read all WARNINGS, CAUTIONS and NOTES in the QIII DC Controller Instruction Book (GEK-45167) prior to installing this Option Kit.*

NOTE: *Read and heed all WARNING, CAUTION, and NOTE labels posted on the equipment prior to installing this Option Kit.*

KIT DESCRIPTION AND CONTENTS

DESCRIPTION:

This Option Kit provides a mixture of analog and digital interfaces between the Control card and the equipment. The optional functions provided by this card are as follows:

- -- Analog (AC or DC) tachometer interface
- -- Analog speed meter driver (analog tachs only)
- -- Digital (square wave) tachometer interface
- -- Process follower (current or voltage reference) interface
- -- RS232C and RS422 serial link interface

CONTENTS:

The kit contains the following components (in addition to these instructions):

Component	Part Number	Qty.
Encoder Process Interface Card	531X134EPRB_G1	1
10PL Conn. Cable (G, C Frames)	36A358208AAG09	1
12PL Conn. Cable (G, C Frames)	36A358208AAG10	1
10PL Conn. Cable (D Frames)	36A358218XAG03	1
12PL Conn. Cable (D Frames)	36A358218XAG05	1
7PL Conn. Cable (All Frames)	36A358208AAG05	1
9PL Conn. Cable (All Frames)	36A358208CXG01	1

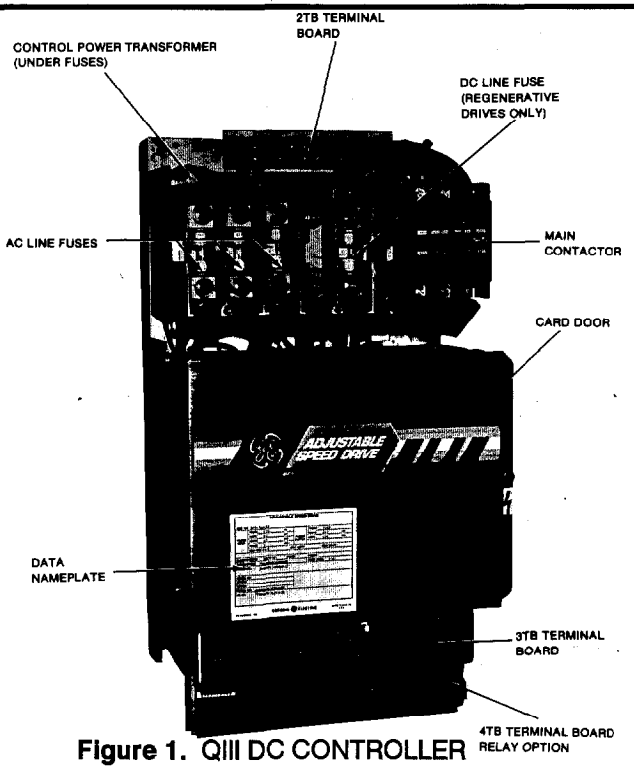


Figure 1. QIII DC CONTROLLER

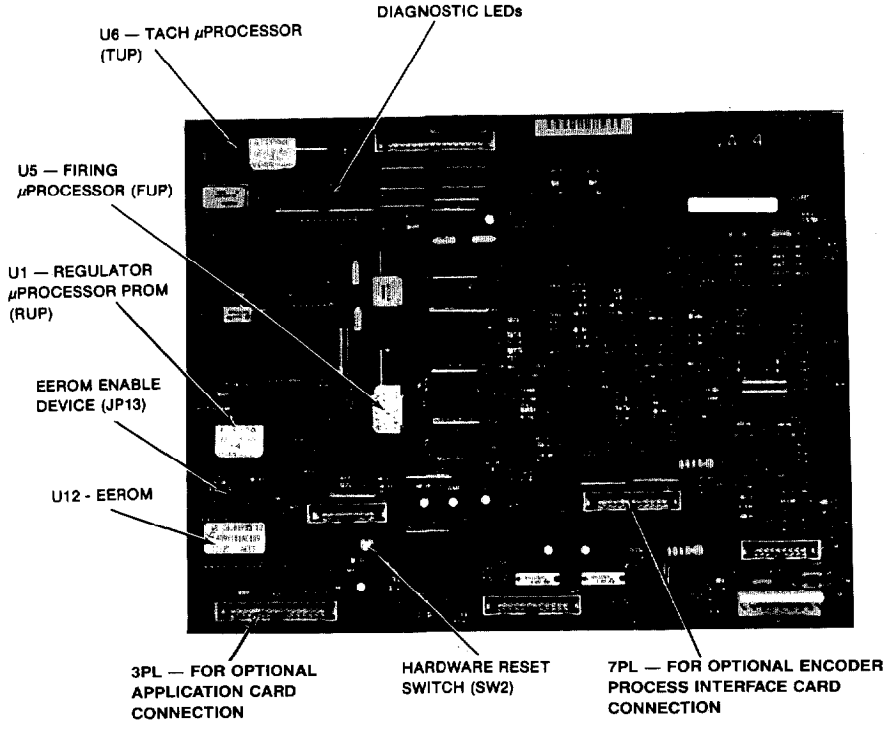


Figure 2. CONTROL CARD

KIT INSTALLATION PROCEDURE

WARNING - SHOCK AND BURN HAZARD:
Always disconnect power to the drive before removing or inserting a printed circuit card. Failure to do so may cause serious injury to personnel and damage to the drive or the driven machinery.

1. Verify that power to the controller that the option kit is to be installed is OFF.

2. Loosen thumbscrew (located on right side) and open the controller card door (see Figure 1).

CAUTION: Treat all cards with static sensitive handling techniques. Use "grounding" strap when changing cards and store cards in the anti-static bags they are shipped in.

3. Mount Encoder Process Interface card to Control card as follows (see Figure 2):
   a. Remove screw (with nylon sheath) from each of the card standoffs (four) located on the right side of the Control card.
   b. Remove Encoder Process Interface card from anti-static bag and orient card with 7PL receptacle towards bottom of card rack.
   c. Position holes in each corner of card over the four card mount standoffs on the Control card.
   d. Secure Encoder Process Interface card to card standoffs by inserting and tightening the screws (with nylon sheaths) removed in step "a".

4. Install 7PL and 12PL ribbon cables on Encoder Process Interface card as follows (see Figure 3):
   a. Orient each ribbon cable to its respective receptacle so that cable extends toward bottom of card rack.
   b. Insert each connector into its respective receptacle on the Encoder Process Interface card and press into fully seated position.

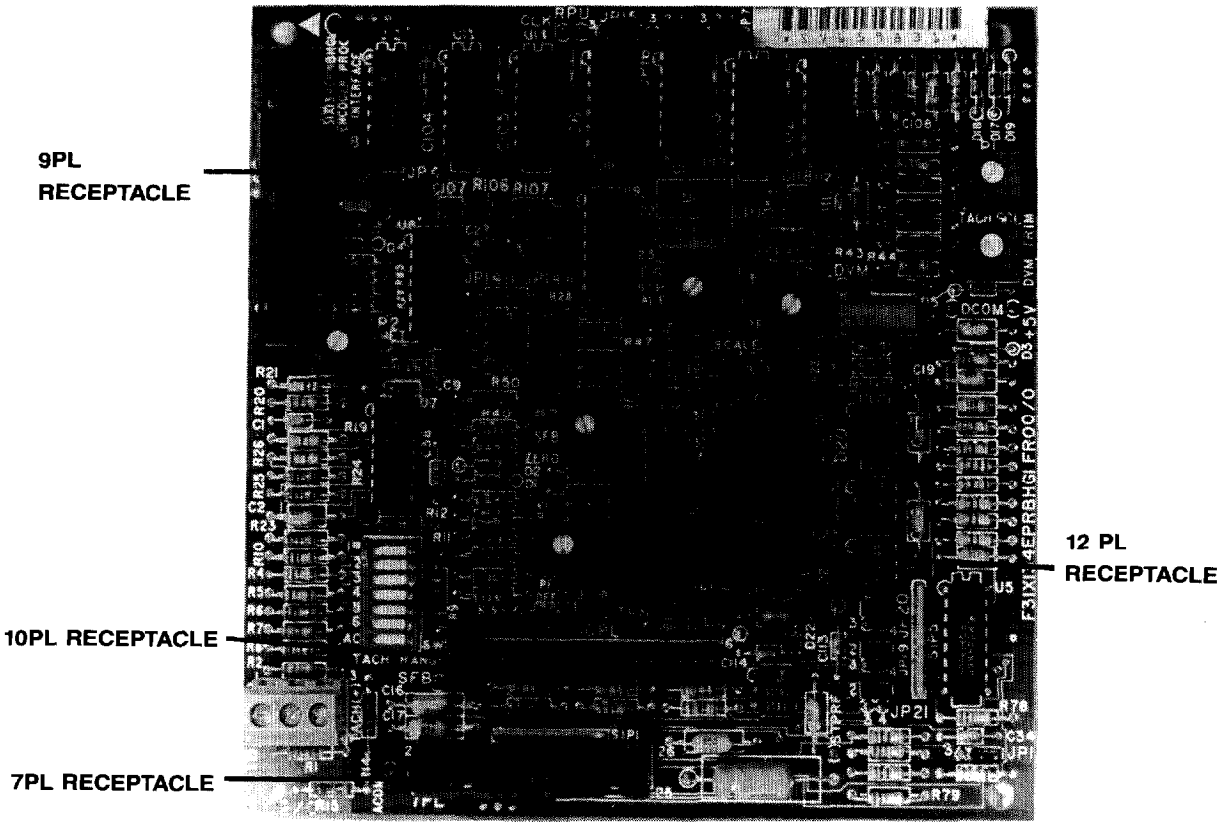


Figure 3. ENCODER PROCESS INTERFACE CARD

5. Orient 10PL ribbon cable to its respective receptacle on the Encoder Process Interface card with the step on the cable connector towards the retaining clip on the receptacle and press the connector into position.

6. Ensure that 7PL cable is not twisted and insert the connector into its receptacle on the Control card and press into fully seated position (see Figure 2).

7. Install 9PL ribbon cable as follows (see Figure 3):
   a. Orient ribbon cable to its receptacle on the Encoder Process Interface card so that cable extends toward left side of the card.
   b. Insert connector into its receptacle on the Encoder Process Interface card and press into fully seated position.
   c. Turn ribbon cable upwards 90 degrees and insert the connector into its receptacle on the Control card and press into fully seated position (see Figure 2).

8. Connect 10PL and 12PL cables to 3TB Terminal Board card as follows (see Figures 4 and 5):

   For "C" and "G" Frame Sizes (see Figure 4):

   a. Verify that cables are not twisted.
   b. Orient the 10PL ribbon cable to its respective receptacle on 3TB with the step on the cable connector towards the retaining clip on the receptacle and press the connector into position.
   c. Insert the connector of 12PL ribbon cable into its receptacle on 3TB and press into fully seated position.

   For "D" Frame Size:

   a. Route 10PL and 12PL ribbon cables through opening in bottom of card rack enclosure.
   b. Make 45 degree fold in each ribbon cable and route cables to the right.
   c. Secure the ribbon cables under the three retainers as shown in Figure 5 (release retainer strap, insert the additional ribbon cables, and reattach retainer strap).
   d. Make a mild 45 degree fold in each ribbon cable in order to route cables downward and then another mild 45 degree fold to route cables to the left towards 3TB (see Figure 5).
   e. Secure the ribbon cables under the appropriate retainers as shown in Figure 5 (release retainer strap, insert the additional ribbon cables, and reattach retainer strap).
   f. Make a 45 degree fold in each ribbon cable when it is positioned at its respective receptacle on 3TB (see Figure 4).
   g. Orient 10PL ribbon cable to its respective receptacle on 3TB with the step on the cable connector towards the retaining clip on the receptacle and press connector into position.
   h. Insert the connector of 12PL ribbon cable into its receptacle on 3TB and press into fully seated position.

9. Set all jumpers and potentiometers on the newly installed Encoder Process Interface card per the Programming Software Package (see Figure 6 for Jumper Location and Table 1 for Jumper Description).

10. Close the controller card door and tighten the thumbscrew (located on right side, see Figure 1).

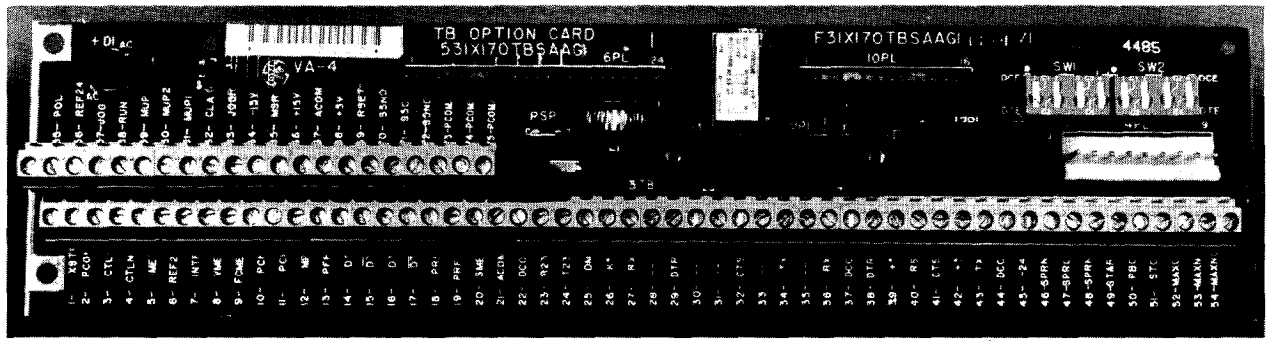


Figure 4. 3TB TERMINAL BOARD CARD

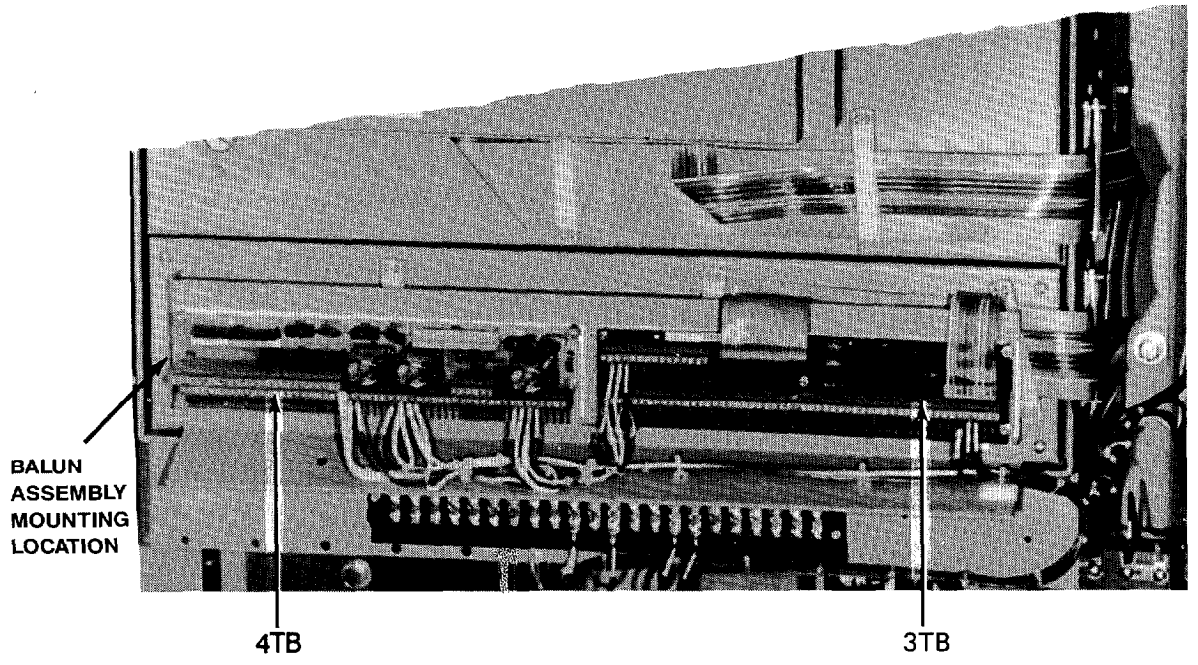


Figure 5. "D" FRAME RIBBON CABLES

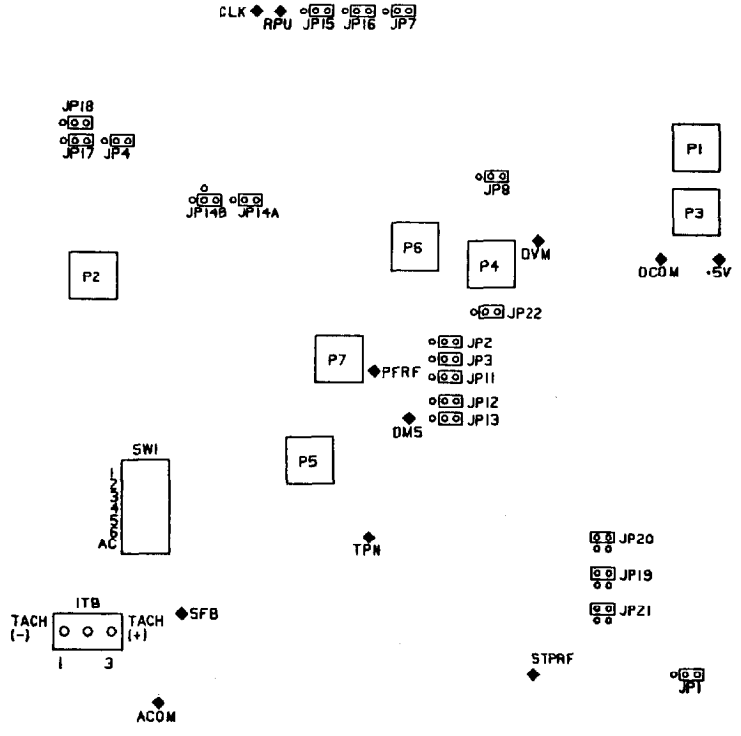


Figure 6. ENCODER PROCESS INTERFACE CARD LAYOUT DIAGRAM

TABLE 1: JUMPER DESCRIPTION - 531X134EPRA_G1 CARDS

Location	Jumper or Pot	Number	Nomenclature	Description
Encoder Process Interface Card	Switch	SW1	Tach Range Select	These switches provide coarse scaling of the tach circuit as follows: DIP switch positions are selected for the legend of the side depressed. To select position 3 ON, depress the switch at position 3 next to the ON legend. **SW1 Position No. "Closed" or "ON"** / **Tach Volt At Maximum Speed DC Tach** / **AC Tach** All "open" or "off" / 10 - 12.5V / 11.1 - 13.9V 1 / 12.5 - 15.5V / 13.9 - 17.2V 2 / 15.5 - 19V / 17.2 - 21.1V 1, 2 / 19 - 22V / 21.1 - 24.4V 3 / 22 - 25V / 24.4 - 27.8V 1, 3 / 25 - 30V / 27.8 - 33.3V 1, 2, 3 / 30 - 37.5V / 33.3 - 41.7V 4 / 37.5 - 46V / 41.7 - 51.1V 1, 2, 4 / 46 - 56V / 51.1 - 62.2V 1, 2, 3, 4 / 56 - 70V / 62.2 - 77.8V 5 / 70 - 87V / 77.8 - 96.7V 2, 3, 5 / 87 - 105V / 96.7 - 117V 1, 2, 4, 5 / 105 - 120V / 117 - 133V 1, 2, 3, 4, 5 / 120 - 144V / 133 - 160V 6 / 144 - 180V / 160 - 200V 1, 2, 4, 6 / 180 - 205V / 200 - 228V 5, 6 / 205 - 250V / 228 - 278V **NOTE:** *SW1 postion 7 must be OFF (or open) for DC tachs or ON (or closed) for AC tachs. Maximum voltages less than 10 volts are not to be used.*
	Pot	P1	TACH SCALE	Used to scale the tach circuit to obtain 4.0 volts at test point SFB at maximum speed.
		P2	SMET	Used to adjust the analog output signal at 3TB20 of the speed metering circuit for 10 volts at maximum motor speed. Maximum current load on this signal is 2mA.
		P3	----	Not Used - Set in full CCW position for proper operation.

Location	Jumper or Pot	Number	Nomenclature	Description
Encoder Process Interface Card (continued)	Pot	P4 P5	PF GAIN PF OFF	These pots allow trimming the gain and offset of the circuit that provides up to 10 volts from 1 to 5, 4 to 20, 5 to 50, or 5 to 65mA transducer inputs. To adjust, start with PF gain at the full CCW position and procede as follows: -- Apply the current reference corresponding to minimum speed. -- Adjust PF OFF for 0 volts at test point PFRF. -- Adjust the current reference corresponding to maximum speed. -- Adjust PF GAIN for 10.0 volts at PFRF. -- Repeat as necessary **NOTE:** *To prevent reversing when the current reference is less than the minimum current, the drive should be set up in a non-reversing mode.*
		P6	----	Not Used - Set in full CCW position for proper operation.
		P7	SFB ZERO	This pot adjusts test point SFB to be zero with zero tach feedback. Set speed reference to zero (jumper MSR or ASR to PCOMX if used) and adjust P7 for zero motor speed when a run is commanded.
	Jumper	JP1		JP1 selects circuitry for the serial link communication mode. **1-2:** Enables RS422 reception. **2-3:** Enables RS232C reception.
		JP2 JP3		JP2 and JP3 select the process follower mode per the following table: **JP2** **JP3** **Range** **1-2:** **1-2:** 5-65ma current loop follower **1-2:** **2-3:** 5-50ma current loop follower **2-3:** **1-2:** 4-20ma current loop follower **2-3:** **2-3:** 1-5ma current loop follower or 10V voltage follower
		JP4		JP4 selects the speed meter (SMET) circuit mode (with analog tachs only). **1-2:** Unipolar speed meter driver mode **2-3:** Bipolar meter driver mode (DC tach only)
		JP7		Not Used - set in 1-2 position for proper operation.
		JP8		JP8 selects the analog tach filter mode. **1-2:** Enables 4ms filter in analog tach feedback (required for AN tach operation). **2-3:** Disables 4ms filter; required for speed regulator performance above 20 radians (DC or Digital Tach only).

Location	Jumper or Pot	Number	Nomenclature	Description
Encoder Process Interface Card (continued)	Jumper	JP11		TXD Mode Select **1-2:** 12PL-9, 12PL-12 RS422 transmitted data mode **2-3:** 12PL-9 RS232 transmitted data mode
		JP12		DTR Mode Select **1-2:** 12PL-8, 12PL-13 RS422 DTR handshake mode **2-3:** 12PL-8 RS232 DTR handshake mode
		JP13		DTR RS232 Logic Polarity **1-2:** 12PL-8 high = drive ready to receive **2-3:** 12PL-8 low = drive ready to receive **NOTE:** *For Series 6 interfaces using the CCM card, set JP1 and JP11 to 1-2 and JP12 and JP13 to 2-3. 12PL-7 should be tied to +5V.*
		JP14A JP14B		CTS Handshake Mode Select **JP14A** / **JP14B** / **Description** **1-2:** / **1-2:** / Enable Differential CTS Handshaking **1-2:** / **2-3:** / Enable Single-Ended CTS Handshaking **2-3:** / **2-4:** / Disable CTS Handshaking
		JP15		Tach Counter Direction Control **1-2:** Normal control by tach direction of rotation. **2-3:** Never
		JP16		Tach A Quad B Enable **1-2:** Normal A quad B Tach operation **2-3:** Never
		JP17		Not Used - set in 1-2 position for proper operation.
		JP18		Not Used - set in 1-2 position for proper operation.
		JP19		Digital Tach Input Signal DTA Scaling **1-2:** 10 Volt Maximum Input **2-3:** 5 Volt Maximum Input **3-4:** 1.5 Volt Maximum Input **1-4:** 15 Volt Maximum Input
		JP20		Digital Tach Input Signal DTB Scaling **1-2:** 10 Volt Maximum Input **2-3:** 5 Volt Maximum Input **3-4:** 1.5 Volt Maximum Input **1-4:** 15 Volt Maximum Input
		JP21		Not Used - set in 1-2 position for proper operation.
		JP22		Voltage Follower Enable **1-2:** Voltage Follower Disabled **2-3:** Voltage Follower Enabled

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